# EVA: Efficient Reinforcement Learning for End-to-End Video Agent

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## Research Questions

- How can an MLLM-based agent autonomously decide when and how to watch a video for efficient understanding?
- How can we train video agents to use planning-before-perception rather than perception-first strategies?
- What training pipeline enables stable reinforcement learning for multi-round video perception and tool usage?

## Core Idea

Traditional video MLLMs passively process entire videos or uniformly sampled frames, leading to inefficient token usage and poor performance on long videos. EVA introduces a planning-before-perception paradigm where the agent first reasons from the text query alone to decide what visual information it needs, then iteratively uses tools to extract targeted frames with adaptive resolution and temporal sampling. The key insight is that most video questions don't require seeing the entire video - by planning first, the agent can allocate visual tokens precisely where needed, avoiding redundant computation while maintaining accuracy.

## Key Formula

### Belief State

$$s_t = \{q, h_t, F_t\}$$

State representation where q is user query, h_t is interleaved text-frame history, F_t is visual evidence from tool calls. This formulation enables the agent to maintain context across multiple rounds of perception and reasoning.

### Composite Reward

$$R(\tau) = w_{acc} r_{acc} + w_{fmt} r_{fmt}$$

Reward function combining accuracy reward (r_acc) and format reward (r_fmt) with weights w_acc and w_fmt. The format reward prevents direct answer guessing without proper tool usage, encouraging visually grounded reasoning.

## Key Figure

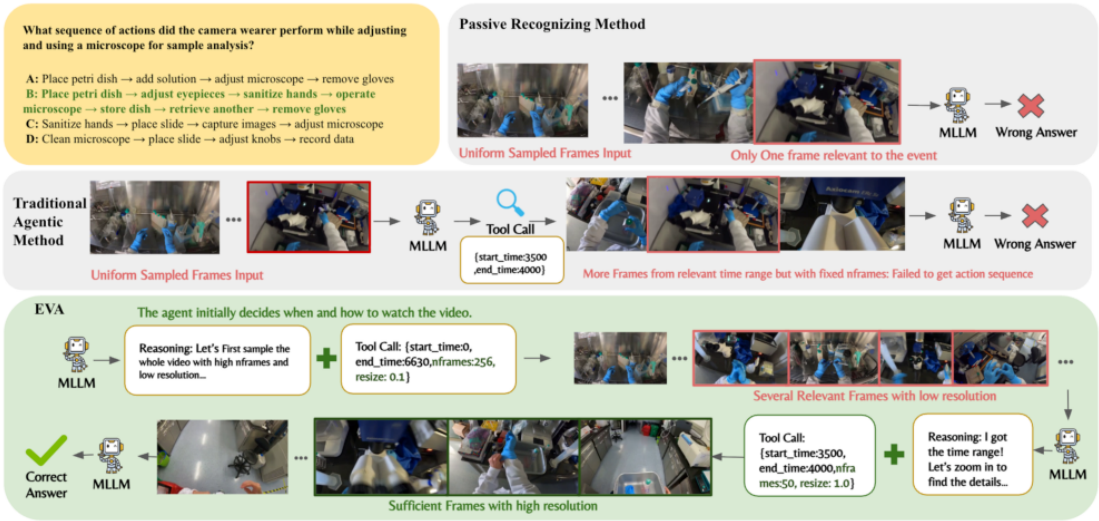


Figure 1. Given a question that requires the MLLM to figure out action sequences from an extremely long video (over 6600 seconds), the traditional uniformed sampling method is limited by the content length of the MLLM, and it is extremely hard for it to sample all the keyframes to answer the question correctly. As for the Traditional Agentic Method, the agent will also be given the uniformly sampled frames along with the video, which already occupy a lot of context. Although the agent can call tools to extract frames from a specific time range, the tool is rigid and the agent cannot adjust the fps and resolution, which leads to potential information loss. However, in EVA, the agent can arrange the tokens wisely. It can first watch the whole video with low resolution and high fps to get an overview of the video without costing too many visual tokens. After it finds the key time range, it will extract frames with high fps and high resolution, which leads to the correct answer.

*Figure 1: Demonstrates EVA's adaptive token allocation strategy compared to traditional uniform sampling and rigid agentic methods for long video understanding.*

## Limitations

- Relies on pre-defined tool interfaces and may struggle with unseen or noisy query distributions
- Current reasoning loop depends on fixed tool schema that may not generalize to diverse video understanding tasks
- Training requires substantial computational resources for the three-stage pipeline with multiple high-quality datasets
- Performance gains are incremental (1-3%) over existing adaptive agent methods, suggesting diminishing returns